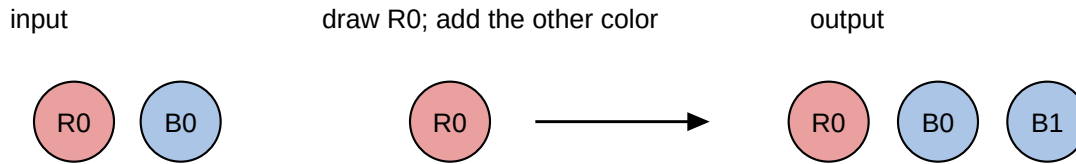


Week 44 / The bag that copies / Grades 2-5

Start with R0 and B0. A random draw gives every counter the same chance. Always return the draw and mix. Any new counter gets the number of the draw that added it.



Problem 1: Compare three rules: add the drawn color, add the other color, or add nothing. Start R0, B0 for each story. Which rule most often gives one red and one blue draw in two draws? Settle it using all the marked stories.

add drawn color	add other color	add nothing
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Week 44 / The bag that copies / Grades 2-5

Use the copying rule: start R0, B0; return each draw, add its own color, and mix. Choose a possible story yourself when designing a bag. For the next random draw, every counter now in the bag has the same chance.

Problem 2: After four copying draws, make the next red draw as likely as possible, as unlikely as possible, and exactly as likely as blue. Give a story and a forecast for each. Can different stories give the same forecast?

red most likely

story: _____

next red forecast

red least likely

story: _____

next red forecast

red and blue tied

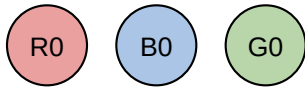
story: _____

next red forecast

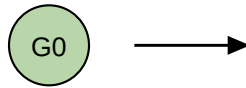
Week 44 / The bag that copies / Grades 4-5

Start R0, B0, G0. Return each draw, add one of its own color, and mix. A new copy gets the number of the draw that added it. Original counters stay in the bag.

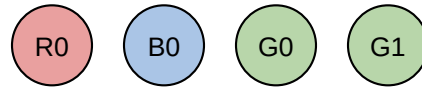
input



draw G0, add G1



output



Problem 3: Use these equally likely marked two-draw stories to find every final color-count bag. Do all bags have the same chance, even though color words may not? Predict the color-count bags after three draws and test your prediction exactly.

 final: _____	 final: _____	 final: _____
 final: _____	 final: _____	 final: _____
 final: _____	 final: _____	 final: _____
 final: _____	 final: _____	 final: _____